

Cosine Attack for Data Reconstruction on the FLICKER Federated Learning Framework

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Abstract

Federated Learning (FL) offers a privacy-preserving approach to collaborative model training, yet faces challenges like class imbalance. The FLICKER framework [3] addresses this by employing the Cheon-Kim-Kim-Song (CKKS) Homomorphic Encryption (HE) scheme, allowing clients to privately balance their datasets. Despite using HE, FLICKER is vulnerable to leakage of client data points. In this work, we identify two significant data reconstruction attacks in the presence of passive adversaries: (i) The dominant client can exploit the shared cosine similarities, and (ii) The server can decrypt buffer vectors and cosine values to reconstruct client data. These two attacks can independently recover the data points of all the clients. We further propose mitigation strategies to enhance FLICKER's privacy guarantees.

Keywords

Federated Learning, Homomorphic Encryption, Data Reconstruction Attacks, Class Imbalance

1 Introduction

FL is a distributed machine learning paradigm designed to enable collaborative model training on decentralized devices without requiring the direct sharing of raw data. This approach is particularly valuable in scenarios where data privacy is of paramount concern, especially in sensitive domains such as healthcare and finance. Although FL is vulnerable in certain scenarios, it remains an attractive choice in most settings [5].

A significant challenge in machine learning, particularly in distributed environments like FL, is class imbalance. When training datasets exhibit skewed class proportions, it can lead to biased predictions and poor generalization in machine learning models. Privacy concerns make detecting and correcting imbalance much more difficult. Sharing class distribution data with a central server could leak sensitive client information. Since clients typically want to keep their datasets confidential, they are limited in what information they can share.

To address this, Guleria et al. [3] propose FLICKER as a privacy preserving framework to address issues related to global class imbalance in FL. The broad outline of FLICKER is as follows:

1. Clients use CKKS homomorphic encryption to securely share local data distributions with the server in a ring network for global imbalance computation.
2. The server computes cosine similarities between each client's local distribution and the global distribution. The client with the highest cosine similarity is selected as the *dominant client*.
3. The dominant client performs local re-balancing through over-sampling minority classes and under-sampling redundant data.

4. The global imbalance is recalculated and the process iterates until the threshold is met or all clients reach their capacity.

Our contribution. We demonstrate two independent data reconstruction attacks on FLICKER. The first attack exploits shared cosine similarities at the dominant client. We call this method a *cosine attack* for data reconstruction. To the best of our knowledge, this is a novel attack which goes beyond FLICKER. This may be of independent interest in broader data reconstruction scenarios. The second attack shows that the server can decrypt buffer vectors and cosine values and use them to reconstruct client data. We also provide recommendations for safe use of FLICKER.

2 Cosine Attack

In the FLICKER resampling stage (Step 3), the dominant client learns the cosine similarity values of other clients w.r.t. its own buffer vectors. Despite using HE, it creates a significant risk for data reconstruction, particularly for low-dimensional data.

Cosine similarity provides directional information about unknown data points. When the dominant client knows multiple vector directions and their angles to a target point, these geometric constraints significantly narrow the possible values in feature space. By intersecting these directional constraints, the client can pinpoint and reconstruct the original data point.

Unlike existing cosine attacks used for membership inference [1], our attack reconstructs complete data points rather than determining membership status. This represents a more severe privacy breach, as the dominant client can potentially recover data points from all clients, completely undermining FLICKER's privacy guarantees. This attack occurs in two phases.

Phase 1: Given buffer points and cosine similarities, infer the direction of intersection.

Phase 2: Using the normalized direction from Phase 1 and the metadata of the dataset, reconstruct the original data point.

2.1 Data Reconstruction

Threat model. An *honest-but-curious* dominant client observes k signed cosine similarities c_i , $1 \leq i \leq k$ between an unknown unit vector $y \in \mathbb{R}^d$ and k known non-zero unit buffer vectors $u_i \in \mathbb{R}^d$. Together, they satisfy the following k equations,

$$c_i = u_i^\top y, \quad \text{where } \|u_i\| = 1.$$

Phase 1. The goal is to recover the *direction* of y (as scale is not identifiable from cosines alone).

Each known vector u_i along with its cosine angle c_i can be used to generate the cone whose apex is at the origin:

$$C(u_i, \theta_i) = \{y \in \mathbb{R}^d : u_i^\top y = \cos \theta_i\}$$

with the axis aligned to u_i and opening angle $\theta_i = \arccos(c_i)$. Hence, the possible surface on which the data point lies is the intersection of all these k cones:

$$\mathcal{F} = \bigcap_{i=1}^k C(u_i, \theta_i) = \{y \in \mathbb{R}^d : u_i^\top y = c_i \forall i\}.$$

The k non-degenerate constraints make $\mathcal{F}$ a $(d - k)$ -dimensional surface. In matrix notation, we can stack the k linear equations into the system

$$Uy = c, \quad \text{where} \\ U = [u_1^\top, \dots, u_k^\top]^\top \in \mathbb{R}^{k \times d}, \quad c = [c_1, \dots, c_k]^\top \in \mathbb{R}^k.$$

Further, if $k = d$, then $\mathcal{F}$ is a straight line passing through origin on which the data point lies (as shown in Figure 1). If $U \in \mathbb{R}^{d \times d}$ is non-singular, we can solve the following linear system to uniquely compute the direction of y :

$$y = U^{-1}c.$$

Complexity: The matrix inversion requires $O(d^3)$ time. Once U^{-1} is calculated, the unknown point can be computed in $O(d^2)$ time. To compute n unknown points, we need $O(d^3 + nd^2)$ time.

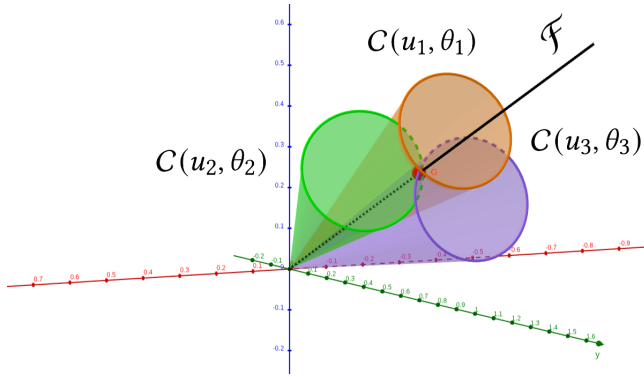


Figure 1: Illustration of intersection of 3 cones in $\mathbb{R}^3$

Phase 2. To pinpoint the exact data point, we use available metadata such as feature types (categorical, ordinal, integer) and feature value ranges (minimum, maximum, non-negative constraints). We illustrate this in the following example.

Example: We demonstrate the cosine attack on the BMI dataset [4]. The dataset includes four columns: Gender, Height, Weight and BMI. The gender is encoded as -1 for females and $+1$ for males. The attack works for different encodings as well. Our implementation is available on GitHub [2].

Phase 1 (Direction Recovery) Let $y = [-1, 151, 64]$ be hidden from the dominant client.

Inputs: Buffer vectors: $x_1 = [-1, 165, 143]$, $x_2 = [1, 197, 149]$, $x_3 = [-1, 158, 149]$ and vector c of cosine similarities of x_i 's w.r.t. y .

Procedure: To find the direction vector, we define the matrix U with $u_i = x_i / \|x_i\|$ and solve for $y = U^{-1}c$.

$$U = \begin{bmatrix} u_1^\top \\ u_2^\top \\ u_3^\top \end{bmatrix} = \begin{bmatrix} -0.0046 & 0.7558 & 0.6550 \\ 0.0040 & 0.7976 & 0.6032 \\ -0.0046 & 0.7276 & 0.6858 \end{bmatrix}, \quad c = \begin{bmatrix} 0.9513 \\ 0.9697 \\ 0.9376 \end{bmatrix}.$$

Output: Direction of intersection $y = (-0.00610, 0.9207, 0.3902)$ and the line of intersection is

$$\mathcal{F}(t) = (t \cdot 0.00610, t \cdot 0.9207, t \cdot 0.3902), \quad t \in \mathbb{R}.$$

Phase 2 (Data Reconstruction)

Inputs: $y = (y_1, y_2, y_3) = (-0.00610, 0.9207, 0.3902)$ and meta-data, specifically gender $g \in \{-1, +1\}$.

Procedure: From the metadata, we know that the first coordinate of the data point has to be $+1$ or -1 . We compute a scaling factor α as $\alpha = \frac{g}{y_1}$ for $g \in \{-1, +1\}$. This scaling ensures that when we adjust the vector's first coordinate, it aligns exactly with the known gender value, anchoring the reconstruction of the point $p = (p_1, p_2, p_3)$. Compute $p = \alpha y$, set $p_1 = g$, $p_2 = \alpha y_2$ and $p_3 = \alpha y_3$. This yields two possible points: $[-1, 151, 64]$ and $[1, -151, -64]$.

Output: Since the second and third coordinates represent height and weight, the reconstructed point is $p = [-1, 151, 64]$.

To avoid the cosine attack, we recommend not using FLICKER for low-dimensional data, as an adversarial client is highly likely to have $k = d$ buffer vectors across multiple rounds of resampling. Thus, making reconstruction from metadata feasible. However, in high-dimensional data having $k = d$ buffer vectors is nearly impossible. Since $\mathcal{F}$ is a $(d - k)$ -dimensional surface, we believe that the constraints imposed by metadata alone may not suffice to recover data points. Thus, FLICKER is safe to use for high-dimension data.

3 Data Leakage at Server

During resampling in FLICKER, the dominant client sends encrypted buffer vectors to the server using the server's public key. An *honest-but-curious* server gets to know the normalized buffer vectors upon decryption. In addition, it can decrypt the cosine similarities encrypted by other clients. Thus, the server can execute the cosine attack described in the previous section to recover all the data points from all the clients.

The cosine attack is possible at the server because the buffer vectors and cosine values are encrypted using the server's public key. To avoid this, we propose the following:

1. **Client-side key generation:** Each client creates its own HE key. During resampling, the dominant client encrypts its buffer points using its public key. This prevents leakage of buffer points at the server.
2. **Server as key manager only:** The server stores the evaluation keys of all the clients, needed for homomorphic operations, but never gains decryption capability.
3. **Round-based key provisioning:** In each round, the server shares evaluation key of the dominant client with other clients to facilitate homomorphic computation of cosine similarity.

The above modifications prevent leakage of buffer points and cosine values at the server. Thus ensuring the privacy guarantees of FLICKER with minimal overhead.

4 Conclusion and Future Work

This paper demonstrates data reconstruction attacks on FLICKER. We have identified two independent vulnerabilities, one at the dominant client and another at the server. To avoid data leakage at the dominant client, we recommend not to use FLICKER for low-dimensional data. We further recommend a patch to avoid data leakage at the server.

This is a work-in-progress. Future work will focus on a more efficient and robust FLICKER framework resistant to the cosine attack that works for low-dimensional data as well.

References

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